Electrical and Computer Engineering

Spoken Language Processing – ENCS5344 Summer – I 2024/2025

Mini-Project

Deadline: Monday 18/8/2025 on ITC

Project Overview:

As you know, this course requires a comprehensive term project that will constitute 15% of your course grade. Since most of these projects require an intensive workload, you can work on this project in groups of **three** (not two and not four!) students.

Project report:

Each group has to submit a short report (2-4 pages long) describing their project in the IEEE transaction letter format (including appendices, figures, references, and everything else you choose to submit). The following is a suggested structure for the final report:

1. Front page (Title, student's names, ID's, etc)

2. Abstract: It should not be more than 200 words

3. Introduction: this section introduces your problem, and the overall plan for approaching your problem

4. Background/Related Work: This section discusses relevant literature for your project

5. Methodology (system description): This section details the framework of your project. Be specific, which means you might want to include equations, figures, plots, etc

6. Experiments and Results: This section begins with what kind of experiments you're doing, what kind of dataset(s) you're using, and what is the way you measure or evaluate your results. It then shows in details the results of your experiments. By details, we mean both quantitative evaluations (show numbers, figures, tables, etc) as well as qualitative results (show images, example results, etc).

7. Conclusion and future work: What have you conclude from the conducted experiments? Suggest future ideas to enhance the results.

8. References: This is absolutely necessary.

IEEE conference paper template is found on the course page at Moodle (itc.birzeit.edu).

Project Idea:

Develop an Automatic Vowel Classification in English Speech Using Formants and MFCC Features with Machine Learning

Dataset:

Use the **Vowel Dataset** or record your own samples.

Option 1: Vowel Dataset

- Contains audio examples and spectrograms of vowels produced by adult male, adult female, and children (ages 3–7).
- Useful for exploring how formants (F1 and F2) vary across speaker age and gender.
- Available at this link: https://personal.utdallas.edu/~assmann/KIDVOW/

Option 2: Record Your Own Dataset

- Record vowels (e.g., heed, hid, hayed, head, had, etc.) from multiple speakers.
- Use Praat or Audacity for labeling and segmentation.

Project Tasks:

Step 1: Preprocessing

- Normalize sampling rate (e.g., 16 kHz)
- Segment vowels manually or automatically
- Apply pre-emphasis and windowing

Step 2: Feature Extraction

A. Formant Extraction

- Use `praat`, `python-speech-features`, or `librosa + LPC` to extract **F1 and F2**
- Plot **F2 vs. F1** vowel space

B. MFCC Extraction

- Use `librosa.feature.mfcc()` to extract **13–20 MFCCs**
- Optionally apply **delta** and **delta-delta** features

Step 3: Machine Learning Models

Use **Scikit-learn** or another library to train classifiers:

- **Model Choices**:
  - Vector Quantization
  - Gaussian Mixture Model (GMM)
  - Support Vector Machine (SVM)
  - Random Forest
  - Logistic Regression
- **Training + Testing**:
  - Use k-fold cross-validation (e.g., 5-fold)
  - Evaluate with **accuracy**, **confusion matrix**, **precision/recall**

Step 4: Evaluation and Comparison

- Compare performance of:
  - Classifier using **Formants (F1, F2) without and with machine learning**
  - Classifier using **MFCC features with machine learning**
- Discuss:
  - Which features worked better and why
  - How performance differs across vowels

Deliverables:

- Python/R code and models
- Visualization:
  - F2 vs F1 vowel plots
  - Confusion matrices
  - MFCC heatmaps (one for each vowel)
- Report including:
  - Methodology
  - Feature analysis
  - Classifier performance
  - Discussion & conclusion

Tools

- **Python** (NumPy, SciPy, librosa, matplotlib, scikit-learn)
- **Praat** (for manual formant extraction or segmentation)

- **Audacity** (for recording)
- **Jupyter Notebook** or **Google Colab**